FastSpeech 2

English and Telugu Assignment - Technical Progress Report

Status as of 8 November 2026

Executive summary

This report documents completed, reproducible work and the current blockers. It is a progress report, not a claim that the assignment is complete. The English Svarah corpus was accessed, exported locally, and audited into duration-targeted train and validation splits. Telugu data have not been obtained or validated. English preparation and forced alignment have now been run, but several utterances failed alignment and English feature extraction/training has not completed. Telugu data are unavailable in this workspace. No checkpoint-backed speech, measured training curves, or variance-adaptor plots are available; none are fabricated here.

Verified English data

Measure	Observed
Source	ai4bharat/Svarah, source split 'test'
Pinned revision	ebbf7777fe771490696a3f7b007097606fa8c924
Export	6,656 WAV utterances; 9.6144 measured audio hours
Training split	3,429 utterances; 5.0007 hours
Validation split	686 utterances; 1.0001 hours
Overlap / speakers	0 shared IDs; Female and Male labels in both splits
Transcript handling	Split audit left transcript strings unchanged. Correctness has not been manually verified.

The audit is recorded in `data/Svarah/splits/audit.json`; the export metadata is in `data/Svarah/export_summary.json`. Raw audio and manifests are local and excluded from version control. The Svarah dataset card identifies CC BY 4.0; see References.

Implementation and reproducibility

The repository vendors the MIT-licensed ming024/FastSpeech2 reference implementation at commit d4e79eb52e8b01d24703b2dfc0385544092958f3. Language-specific configuration and wrapper scripts, a dataset exporter, and a deterministic duration-based manifest audit are present. The English split uses seed 42. The audit checks audio durations, duplicate IDs, missing files/transcripts, target duration, and train/validation ID overlap.

The project-local Python environment reports PyTorch 2.14.1+cpu and no CUDA device. English audio was normalized to mono 22.05 kHz PCM16, transcripts were normalized for the configured English cleaner, a corpus-specific 6,456-word G2P lexicon was generated, and the official English US ARPA MFA acoustic model was used. Alignment produced 3,367 of 3,429 training TextGrids and 667 of 686 validation TextGrids. The missing training items total about 8.8 minutes, leaving about 4.85 aligned hours and outside the five-minute duration tolerance; validation retains about 57.8 aligned minutes. Alignment failures require investigation or a documented replacement split before feature extraction can complete. MFA validation was skipped because its temporary training pass was too slow. No model training or inference has completed.

Assignment status

Requirement	Status	Evidence / remaining work
English corpus and 5 h / 1 h split	Complete	Verified Svarah export and disjoint split audit.
Telugu corpus and split	Not started	A public mirror has conflicting language metadata; original IIT Madras access/license process remains unresolved. Do not assume the mirror is Telugu.
English normalization, lexicon, forced alignment	Partial	English MFA TextGrids exist, but 62 train and 19 validation utterances failed; train aligned duration is outside tolerance. Telugu resources remain unvalidated.
FastSpeech 2 training and checkpoints	Not started	Train separate language models; record configuration, hardware, and actual training/validation losses.
Inference, five test cases per language, plots	Not started	Requires trained checkpoints; preserve assignment test strings exactly and render duration, pitch, and energy plots.
Progress report and public repository	Available	This progress report is not a final experimental report. Public code repository: https://github.com/mansoorakhanam27/Fast2Speech. Local audio, environments, credentials, and checkpoints are excluded.

Next steps

1. Obtain the original Telugu corpus through an authorized route, review its license, and verify transcript language, audio, speakers, and duration before export. 2. Install and validate MFA and language-specific pronunciation resources. 3. Run preprocessing and forced alignment for each language, manually inspect alignment samples, and extract mel, pitch, energy, and duration targets. 4. Train both models on a CUDA-capable machine, preserving logs and checkpoints. 5. Synthesize the exact assignment test sentences, generate the required plots, and complete the final report from measured results. 6. Update the existing public GitHub repository with the final verified code and report; do not publish restricted audio or credentials.

Reproduce the completed split audit

From the project root in PowerShell:

```
python tools\audit_manifests.py --manifest data\Svarah\manifest.jsonl --audio-root data\Svarah --output-dir data\Svarah\splits --language English
```

The workspace is local at `C:\Users\manso\OneDrive\Documents\ChatGPT\Fast2Speech`. Public repository: https://github.com/mansoorakhanam27/Fast2Speech. Raw dataset audio, credentials, and local environments are excluded.

References

1. FastSpeech 2 assignment handout supplied by the student, *FastSpeech2_Assignment.pdf*.

2. ming024/FastSpeech2 reference implementation, commit d4e79eb52e8b01d24703b2dfc0385544092958f3. The upstream README reports under 10,000 steps (under one hour on a GTX 1080 Ti) to generate acceptable-quality samples; this is not a runtime guarantee for this project or its configurations. github.com/ming024/FastSpeech2.

3. AI4Bharat, Svarah dataset card and terms. huggingface.co/datasets/ai4bharat/Svarah.

4. SPRINGLab IndicTTS Telugu dataset card. Its description includes conflicting language information; use only after verification. huggingface.co/datasets/SPRINGLab/IndicTTS_Telugu.

Limitations: This is a technical progress report, not a completed assignment report. Alignment success does not prove transcript correctness or manually verified phone boundaries. There are no trained checkpoints, model metrics, or generated samples yet.